2026 04.28 → 06.19

数据科学与大数据技术的数学基础

Mathematical Foundations of Data Science

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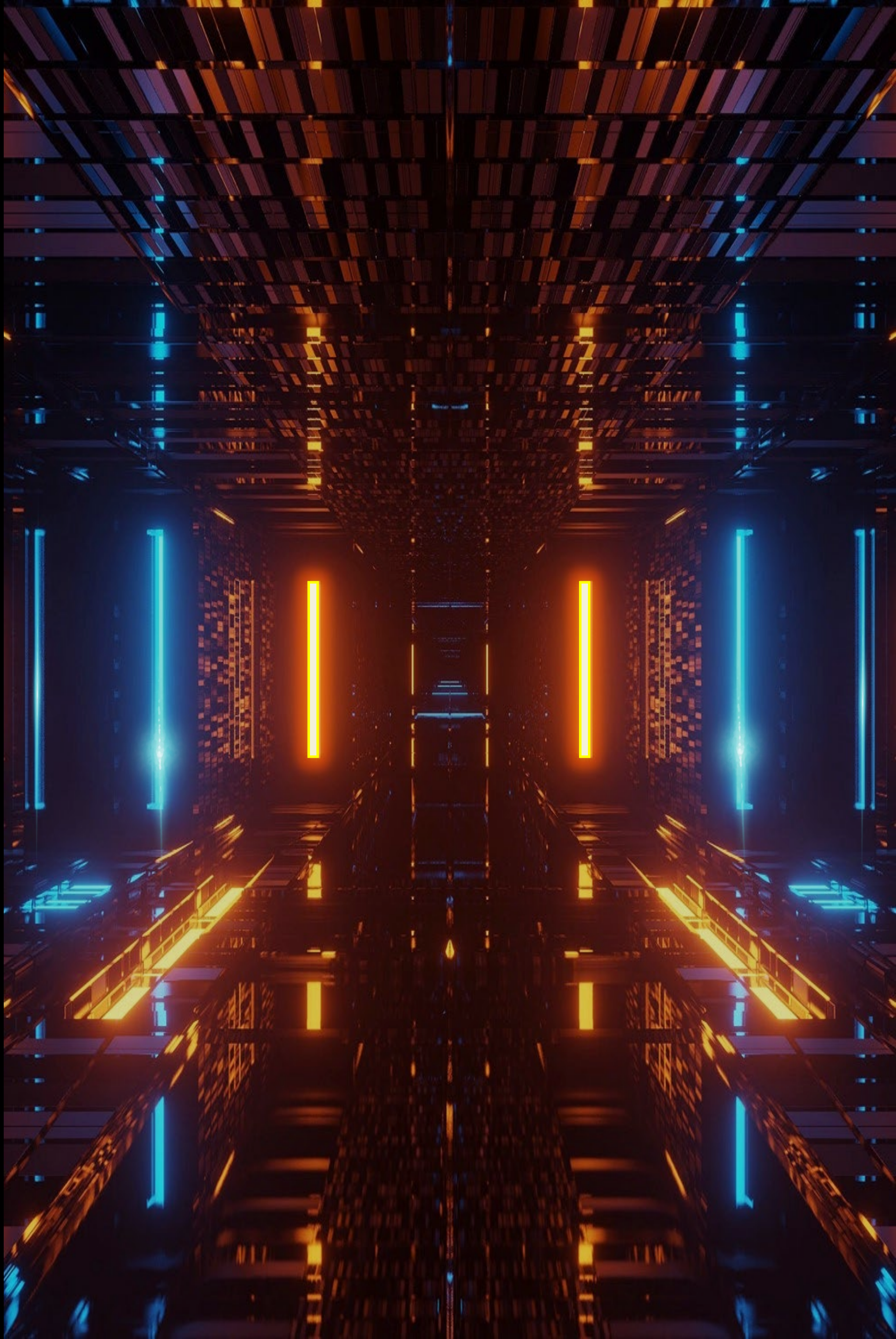

01

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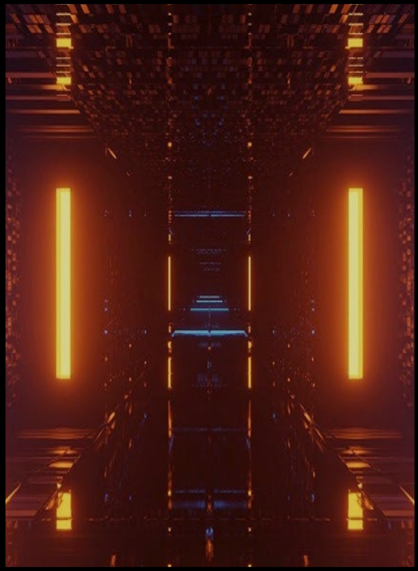
Mathematical Foundations of Data Science

/05 奇异值分解

Singular Value Decomposition(SVD)



PS Find the eigenvalues and eigenvectors

1. Calculate the eigenvalues and eigenvectors of **A** = $\begin{bmatrix} -\mathbf{1} & \mathbf{4} \\ -\mathbf{2} & \mathbf{5} \end{bmatrix}$

① find eigenvalues

Let the determinant $|\boldsymbol{\lambda I} - \boldsymbol{A}| = 0$ (the eigenvalue definition $\boldsymbol{Ax} = \boldsymbol{\lambda x}$, after shifting $(A - \lambda I)x = 0$, because the eigenvector x is the non- zero solution of (A-λE) x =0, and the determinant $|\lambda I - A|$=0 is a necessary and sufficient condition for homogeneous linear equations to have non-zero solutions), and I is the identity matrix, then

$$|\boldsymbol{\lambda I} - \boldsymbol{A}| = \begin{vmatrix} \lambda + 1 & -4 \\ 2 & \lambda - 5 \end{vmatrix} = \lambda^2 - 4\lambda + 3 = 0 \rightarrow \lambda_1 = 3; \lambda_2 = 1$$

The line with the smaller slope λ=1 means that the length of the vector on this line has not changed; the other line corresponds to λ=3, which means that the length of the vector is extended by 3 times.

PS Find the eigenvalues and eigenvectors

②Find the corresponding eigenvector

$\boldsymbol{When\ \lambda = 1, let\ (\lambda I - A)x = 0}$

$$\lambda I - \mathrm{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -1 & 4 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} 2 & -4 \\ 2 & -4 \end{bmatrix}$$

Let $(\lambda I - A)x = 0, x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, corresponding to this straight line with smaller slope, the length of all vectors on the line does not change.

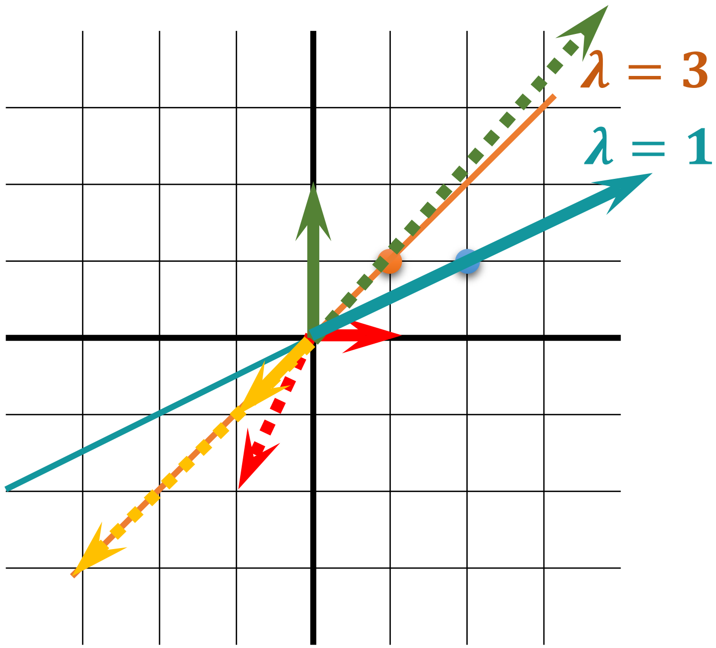
When λ=3, let $(\lambda I - A)x = 0$

Let $(\lambda I - A)x = 0, x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, corresponding to the straight line with larger slope, the length of all vectors on the line is enlarged by 3 times.

Note: In PCA, the eigenvector needs to be converted into a **unit vector**, and a **non-zero vector divided by its modulus** can obtain the required unit vector

PS Find the eigenvalues and eigenvectors

Suppose A =$\begin{bmatrix}-1 & 4\\-2 & 5\end{bmatrix}$is the transformation matrix, verify the geometric meaning: known $Ax = \lambda x$(x are two eigenvectors respectively)



① For the vector in the same direction as the eigenvector of the transformation matrix, when **left multiply** vector by the transformation matrix, the direction remains unchanged, and the scaling size is equal to the eigenvalue that corresponding to the eigenvector.

②The covariance matrix is introduced in PCA, then all the elements in the C matrix are variance and covariance, and then the larger eigenvalue of C means that the eigenvector that can magnify the variance is selected.

In addition, **the covariance matrix is a symmetrical square matrix**, and the eigenvectors corresponding to different eigenvalues are orthogonal and linearly independent. Does it look more like a set of new coordinate axes?

Finally, the centered sample point matrix is multiplied by the normalized eigenvector selected for dimensionality reduction. From each row, it is the definition of vector point multiplication. From a geometric point of view, vector point multiplication is the process of projection.

While $R = \begin{bmatrix}1\\0\end{bmatrix}$, $AR = \begin{bmatrix}-1 & 4\\-2 & 5\end{bmatrix} \times \begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}-1\\-2\end{bmatrix}$

While $G = \begin{bmatrix}0\\2\end{bmatrix}$, $AR = \begin{bmatrix}-1 & 4\\-2 & 5\end{bmatrix} \times \begin{bmatrix}0\\2\end{bmatrix} = \begin{bmatrix}8\\10\end{bmatrix}$

While $B = \begin{bmatrix}4\\2\end{bmatrix}$, $AR = \begin{bmatrix}-1 & 4\\-2 & 5\end{bmatrix} \times \begin{bmatrix}4\\2\end{bmatrix} = \begin{bmatrix}4\\2\end{bmatrix} = 1 \times \begin{bmatrix}4\\2\end{bmatrix}$

While $Y = \begin{bmatrix}-1\\-1\end{bmatrix}$, $AR = \begin{bmatrix}-1 & 4\\-2 & 5\end{bmatrix} \times \begin{bmatrix}-1\\-1\end{bmatrix} = \begin{bmatrix}-3\\-3\end{bmatrix} = 3 \times \begin{bmatrix}-1\\-1\end{bmatrix}$

PS Eigenvalue Decomposition

Matrix multiplication corresponds to transformation (don't forget, left multiply vector by the transformation matrix), which is to **turn any vector into a new vector with a different direction and a different length**. During this change process, the original vector mainly undergoes **rotation and stretching changes**.

Which vectors **only scale**? **Vector in the same direction as the eigenvectors of the transformation matrix**.

What about the **scaling ratio**? **Eigenvalues!**

Note: The vectors on the line where the eigenvectors are located are all eigenvectors. Such straight line may be more than one. **We call the set of straight lines corresponding to all the eigenvectors of the matrix A the feature space**

PS Eigenvalue Decomposition

Eigenvalue Decomposition：

Decompose the matrix A into：

$$A = P\Lambda P^{-1}$$

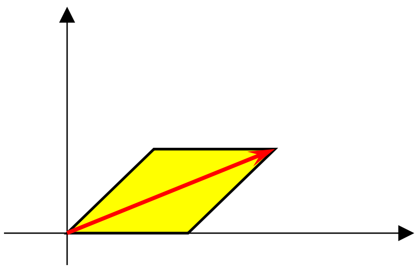
$$x' = \frac{x}{\|x\|}$$

P is a matrix composed of the **eigenvectors** of this matrix A

Λ is a **diagonal matrix composed of eigenvalues**, the eigenvalues are arranged **from large to small**, and the eigenvariables corresponding to these eigenvalues describe the direction of change of this matrix (arranged from major changes to minor changes)

But: only square matrices have eigenvalues and eigenvectors

For **real symmetric matrices or Hermitian matrices**, the eigenvectors corresponding to different eigenvalues must be **orthogonal** (perpendicular to each other)



5.1 From PCA to SVD

Eigenvalue decomposition: The matrix in both eigenvalue decomposition and singular value decomposition can be regarded as a **transformation**. Eigenvalue decomposition can be used **to find vectors with invariant direction (only scaling changes)**, and can also be used for **PCA** to find the projection that maximizes the sample variance. Direction. PCA can be used for **feature dimensionality** reduction, data compression, and turning features into uncorrelated variables.

In the data compression process, we choose the k dimension with the largest eigenvalue. If it is required to retain N% of the amount of information, then calculate the number k of eigenvalues that need to be retained according to the **cumulative variance percent formula**. Also, it is important to note that eigenvalue decomposition can only be used for **square matrices**.

Singular value decomposition: It simply serves to **extract the key features in a more complex matrix** for **simplification** purposes. In contrast, **eigendecomposition is only applicable to square matrices**, while singular value decomposition is applicable to **real matrices**. For this reason, it also has a wide range of applications, such as: LSA (implicit semantic analysis), recommender systems, feature compression (or data dimensionality reduction), PCA (principal component analysis), model parameterization, etc.

5.1 From PCA to SVD

Both eigendecomposition and singular value decomposition are to find a special set of bases for a matrix (linear transformation)

Eigen decomposition

For **square matrix**, a basis consisting of **eigenvectors**, the linear transformation has only a **scaling** effect.

$$\boldsymbol{A} = \boldsymbol{W\Sigma W^{-1}}$$

$\boldsymbol{A}$: **square matrix** of order $\boldsymbol{n}$
$\boldsymbol{\Sigma}$:$n \times n$-dimensional matrix with n **eigenvalues** on the diagonal
W: A unitary matrix, where n eigenvectors of W are standard orthogonal bases , $W^TW = I, W^T = W^{-1}$, The 2-norm of the eigenvector (the modulus of the vector) = 1
(The norm of the matrix is different, refer to the column mode and spectral mode)

Singular value decomposition, SVD

For **real matrices**, singular value decomposition is the decomposition of a matrix into the product of **3 simple matrices**, and the effects of **linear transformations such as rotation and scaling** are represented by these simple matrices respectively.

$$\boldsymbol{A} = \boldsymbol{U\Sigma V^T}$$

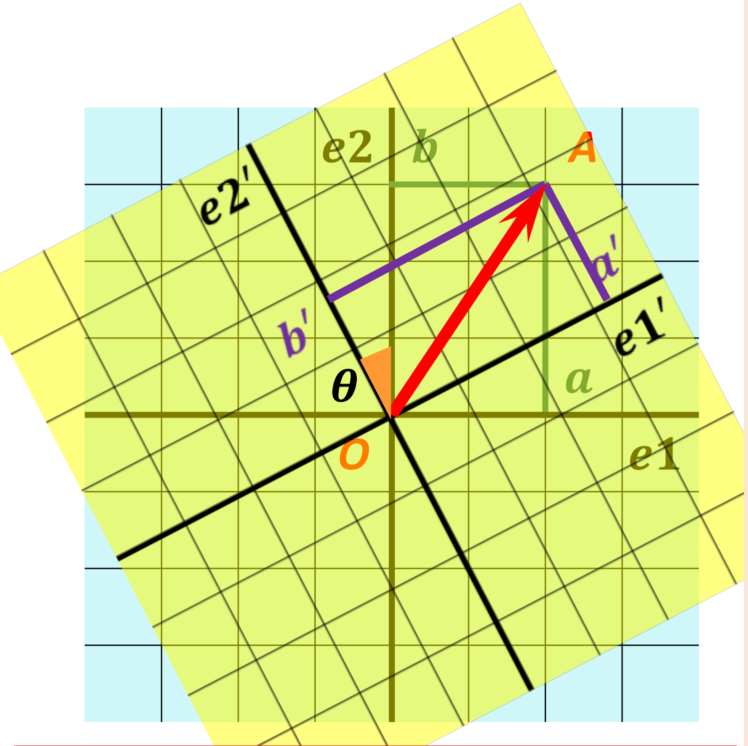
U:$m \times m$-dimensional matrix
Σ:$m \times n$-dimensional matrix with all 0 except for the elements on the main diagonal, and each element on the main diagonal is a singular value
V: $n \times n$ -dimensional matrix
U and V:both are unitary matrix, that satisfy the condition $U^TU = I, V^TV = I$

A **unitary** matrix: any two columns are orthogonal, and each column has length 1 (think of unit-length eigenvectors).

PS Matrix transformation-rotation

Left multiplied vector by a orthogonal matrix
= performing an orthogonal transformation on the vector

①Does orthogonal transformation change the dimensions of vectors?
②Will the variables be rotated?



Assume that the coordinates of **OA** in the standard coordinate system are $(1,1)^T$, and the coordinate axis is rotated 60° counterclockwise to find the new vertex coordinates

1. Suppose a vector $\boldsymbol{OA}$ in two-dimensional space
2. Its coordinates in the standard coordinate system e1、e2 are $(a,b)^T$
3. The coordinate system rotates counterclockwise by θ angle to e1′,e2′. At this time, the coordinates are $(a',b')^T$
4. There is an orthogonal matrix U such that $(a',b')^T = U(a,b)^T$
5. $\boldsymbol{OA}$ has not been stretched, nor has it changed its actual spatial position, but its coordinates have changed.
6. U: $\boldsymbol{x} = \begin{bmatrix} a \\ b \end{bmatrix}, a' = \boldsymbol{x} \cdot e1' = e1'^T \boldsymbol{x},\ b' = \boldsymbol{x} \cdot e2' = e2'^T \boldsymbol{x}$

a', b' are actually the projection of $\boldsymbol{x}$ on the new coordinate system e1′、e2′

7. $\begin{bmatrix} a' \\ b' \end{bmatrix} = \begin{bmatrix} e1'^T \\ e2'^T \end{bmatrix} \boldsymbol{x}$, while $e1' = \begin{bmatrix} \cos\theta \\ sin\theta \end{bmatrix}, e2' = \begin{bmatrix} -sin\theta \\ cos\theta \end{bmatrix}$

$U = \begin{bmatrix} \cos\theta & sin\theta \\ -sin\theta & \cos\theta \end{bmatrix}$(Anticlockwise rotation)

5.2 Singular value decomposition, SVD

Matrix A : $m * n$

$$A = U\Sigma V^T$$

$A^T A$ is a real symmetric matrix and a square matrix

(Rotation transformation)
Orthogonal matrix U is m order matrix

(Scaling transformation)
The diagonal matrix $\sum$ is $m * n$ order matrix

(Rotation transformation)
Orthogonal matrix V is n order matrix

$U^T U = U U^T = I$, unitary matrix

If the unitary matrix U is an n-order square matrix, then the column vectors and row vectors of U respectively constitute a set of **orthonormal basis** on the inner product space C^n

$$AA^T = U\sum^2 U^T$$

Then perform eigenvalue decomposition, the eigenvectors that satisfy $(AA^T)u_i = \lambda_i u_i$ form U

The diagonal of the singular value matrix $\sum$ is the singular value, and the other positions are 0, $\boldsymbol{\sigma_i} = \sqrt{\boldsymbol{\lambda_i}}$

$\boldsymbol{\lambda}$ is the eigenvalue of $\boldsymbol{A^T A}$

Eigenvalue decomposition,
$\boldsymbol{A' = W'\Sigma' W'^{-1}}$,
$\boldsymbol{W'}$: eigenvector, $\boldsymbol{\Sigma'}$: diagonal matrix composed of eigenvalues

$$A^T A = V\sum^2 V^T$$

Then perform eigenvalue decomposition, the eigenvectors that satisfy $(A^T A)v_i = \lambda_i v_i$ form V

计算：

$$A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} = U\Sigma V^T$$

In Singular Value Decomposition (SVD), singular values are not obtained by simply taking the square root of eigenvalues directly in the way you might initially think, and **eigenvalues in the relevant context for SVD cannot be negative**. Here's a detailed explanation:

1. SVD basics and the role of eigenvalues

•**SVD formula**: For a matrix A $(m * n)$, its SVD is given by $A = U\Sigma V^T$, where $U_{m\times m}$ and $V_{n\times n}$ are orthogonal matrices, and $\Sigma_{m\times n}$ is a diagonal matrix with non - negative diagonal elements $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0 (r = rank(A))$, and the remaining diagonal elements are 0.

•**Connection to eigenvalues**: The singular values of *A* are related to the eigenvalues of the matrices A^TA and AA^T. Specifically, the non - zero singular values of *A* are the square roots of the non - zero eigenvalues of both A^TA and AA^T. That is, if λ is a non - zero eigenvalue of A^TA (or AA^T), then the corresponding singular value $\sigma = \sqrt{\lambda}$.

2. Why eigenvalues of $A^T A$ and AA^T are non - negative

•**For any vector** *x*: Consider the quadratic form $x^T(A^T A)x = (Ax)^T(Ax)$. By the property of the dot product, $(Ax)^T(Ax) = \|Ax\|^2$, where $\|Ax\|$ is the Euclidean norm of the vector *Ax*. Since the square of the Euclidean norm of a vector is always non - negative, i.e., $\|Ax\|^2 > 0$ for any vector *x*, the quadratic form $x^T(A^T A)x \geq 0$ for all *x*.

•**Definition of positive semi - definite matrices**: A matrix *M* is called positive semi - definite if $x^T Mx \geq 0$ for all vectors *x*. So, $A^T A$ is a positive semi - definite matrix. Similarly, AA^T is also a positive semi - definite matrix because for any vector *y*, $y^T(AA^T)y = (A^T y)^T(A^T y) = \|A^T y\|^2 \geq 0$.

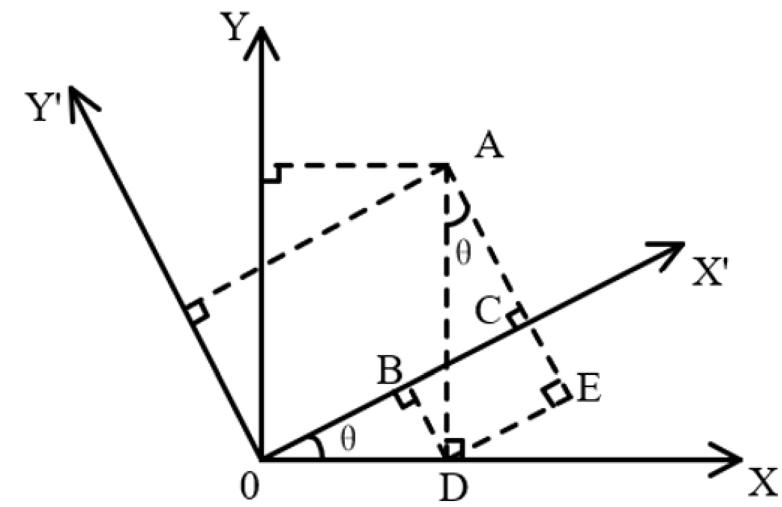
•**Eigenvalues of positive semi - definite matrices**: One of the important properties of positive semi - definite matrices is that all their eigenvalues are non - negative. This can be shown as follows: Let *λ* be an eigenvalue of $A^T A$ and *x* be the corresponding non - zero eigenvector, so $A^T Ax = \lambda x$. Then $x^T(A^T A)x = x^T(\lambda x) = \lambda x^T x$. Since $x^T(A^T A)x = \|Ax\|^2 \geq 0$ and $x^T x > 0$ (because *x* is non - zero), we have *λ*≥0. The same argument holds for the eigenvalues of AA^T.

In conclusion, when calculating singular values from eigenvalues in SVD, **the eigenvalues of $\boldsymbol{A^T A}$ and $\boldsymbol{AA^T}$ are non - negative**, so we can take their square roots to obtain the non - negative singular values.

5.2 Singular value decomposition, SVD

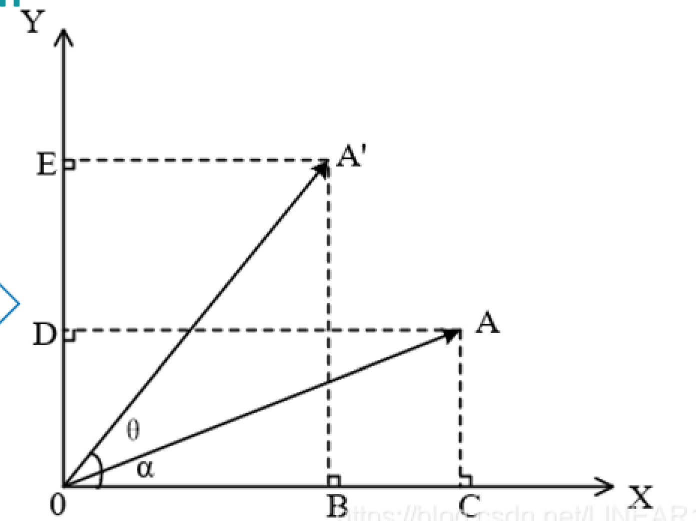
Visualization of the SVD algorithm: the decomposition process of the transformation matrix from the perspective of vectors

The world coordinates of point A after rotation = rotation matrix × the world coordinates of point A before rotation



$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\theta = -\theta$$



$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

The coordinate transformation of a static point A in different coordinate systems
(the point remains the same, but the coordinate system changes)

The world coordinate transformation of a moving point A before and after rotation
(the coordinate system remains the same, but the point changes)

The camera coordinates of point A after transformation = inverse of the rotation matrix × the world coordinates of point A before transformation

5.2 Singular value decomposition, SVD

$A=\begin{bmatrix}3 & 0\\ 4 & 5\end{bmatrix}$ $A^T=\begin{bmatrix}3 & 4\\ 0 & 5\end{bmatrix}$

$AA^T=\begin{bmatrix}9 & 12\\ 12 & 41\end{bmatrix}$ $\begin{vmatrix}\lambda-9 & -12\\ -12 & \lambda-41\end{vmatrix}=0$ $(\lambda-9)(\lambda-41)-144=0$

So $\lambda_1=45$, $\lambda_2=5$

The eigenvector corresponding to $\lambda_1=45$ is $\begin{bmatrix}\frac{1}{\sqrt{10}}\\ \frac{3}{\sqrt{10}}\end{bmatrix}$

The eigenvector corresponding to $\lambda_2=5$ is $\begin{bmatrix}-\frac{3}{\sqrt{10}}\\ \frac{1}{\sqrt{10}}\end{bmatrix}$

Orthonormal basis:

① Orthogonalization

② Column vector normalization

$U=\begin{bmatrix}\frac{1}{\sqrt{10}} & -\frac{3}{\sqrt{10}}\\ \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}}\end{bmatrix}$

5.2 Singular value decomposition, SVD

$A=\begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$ $A^T=\begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$

$A^TA=\begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$ $\begin{vmatrix} \lambda - 25 & -20 \\ -20 & \lambda - 25 \end{vmatrix}=0$ $(\lambda-25)(\lambda-25)-400=0$

So $\lambda_1=45$，$\lambda_2=5$

The eigenvector corresponding to $\lambda_1=45$ is $\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$

The eigenvector corresponding to $\lambda_2=5$ is $\begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$

$$V=\begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

5.2 Singular value decomposition, SVD

$A=\begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$ $A^T=\begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$

$A^TA=\begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$ $\begin{vmatrix} \lambda-25 & -20 \\ -20 & \lambda-25 \end{vmatrix}=0$ $(\lambda-25)(\lambda-25)-400=0$

So $\lambda_1=45$， $\lambda_2=5$

$\sigma_1=\sqrt{\lambda_1}=3\sqrt{5}$ $\sigma_2=\sqrt{\lambda_2}=\sqrt{5}$ $\Sigma=\begin{bmatrix} 3\sqrt{5} & 0 \\ 0 & \sqrt{5} \end{bmatrix}$

$$\begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}=\begin{bmatrix} \frac{1}{\sqrt{10}} & -\frac{3}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix}\begin{bmatrix} 3\sqrt{5} & 0 \\ 0 & \sqrt{5} \end{bmatrix}\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

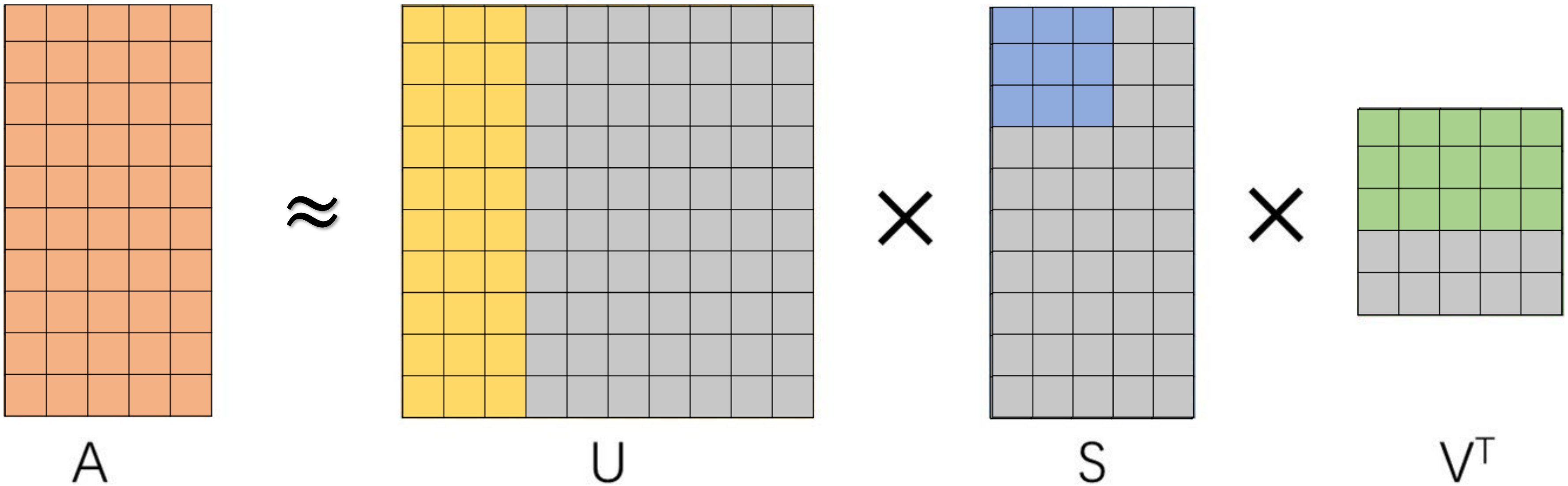
5.3 Application of SVD

1. SVD for image compression

SVD is a commonly used data compression algorithm. In previous courses, we have learned how to compress a large matrix into a form of multiplying several small matrices.

Perhaps you may be wondering why a $m \times n$ matrix calculation SVD decomposition can achieve data compression, after all, whether m is larger or n is larger, there is a square matrix larger than A in the decomposition result.

The image compression of SVD starts after decomposing three matrices, starting from the singular value matrix.

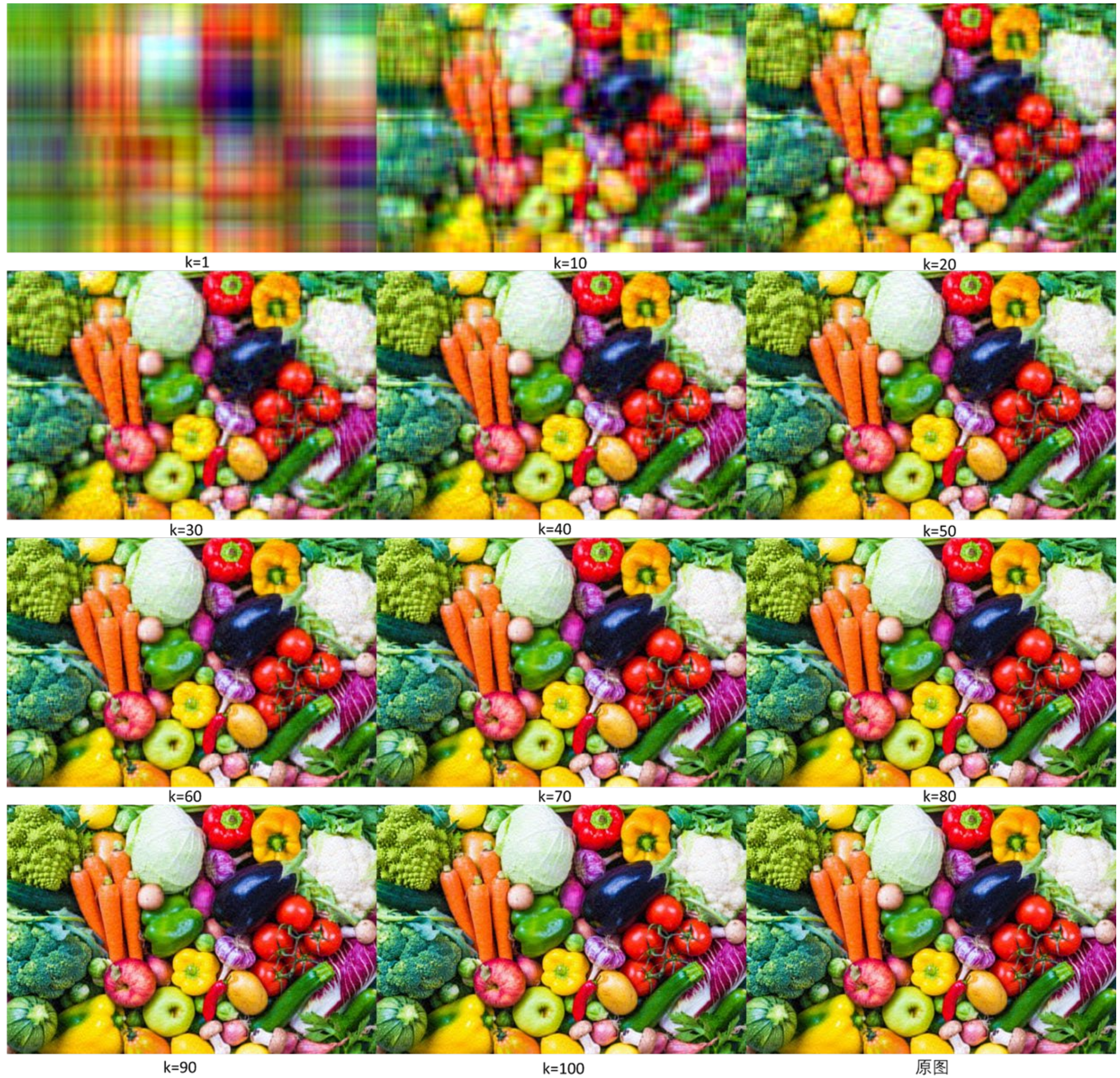


After compression, A is still the original size, but the matrix value has changed

5.3 Application of SVD

1. SVD for image compression

It can be seen that when SVD is used for image compression, **it deletes some small singular values**, which affects the color of all points in the image, cross-color, instead of simply retaining a matrix element value every other bit for down-sampling .



5.3 Application of SVD

1.Data Dimensionality Reduction and Compression

By retaining the singular vectors corresponding to the top k largest singular values, the original matrix can be approximately reconstructed, achieving low-rank approximation. For example, in image compression, SVD reduces data volume by preserving the dominant singular values while retaining key visual information.

2.Noise Removal and Feature Extraction

By ignoring smaller singular values (which typically correspond to noise), a cleaner signal or dataset can be reconstructed. In speech signal processing, SVD effectively separates the signal from background noise.

3.Recommender Systems and Collaborative Filtering

The SVD decomposition of a user-item rating matrix can predict missing ratings and uncover latent user preferences. For instance, Netflix's recommendation algorithm is based on this technique.

4.Latent Semantic Analysis (LSA)

In text mining, SVD decomposition of the document-term matrix captures semantic relationships, enabling synonym mapping and topic modeling.

5.Matrix Pseudoinverse Calculation

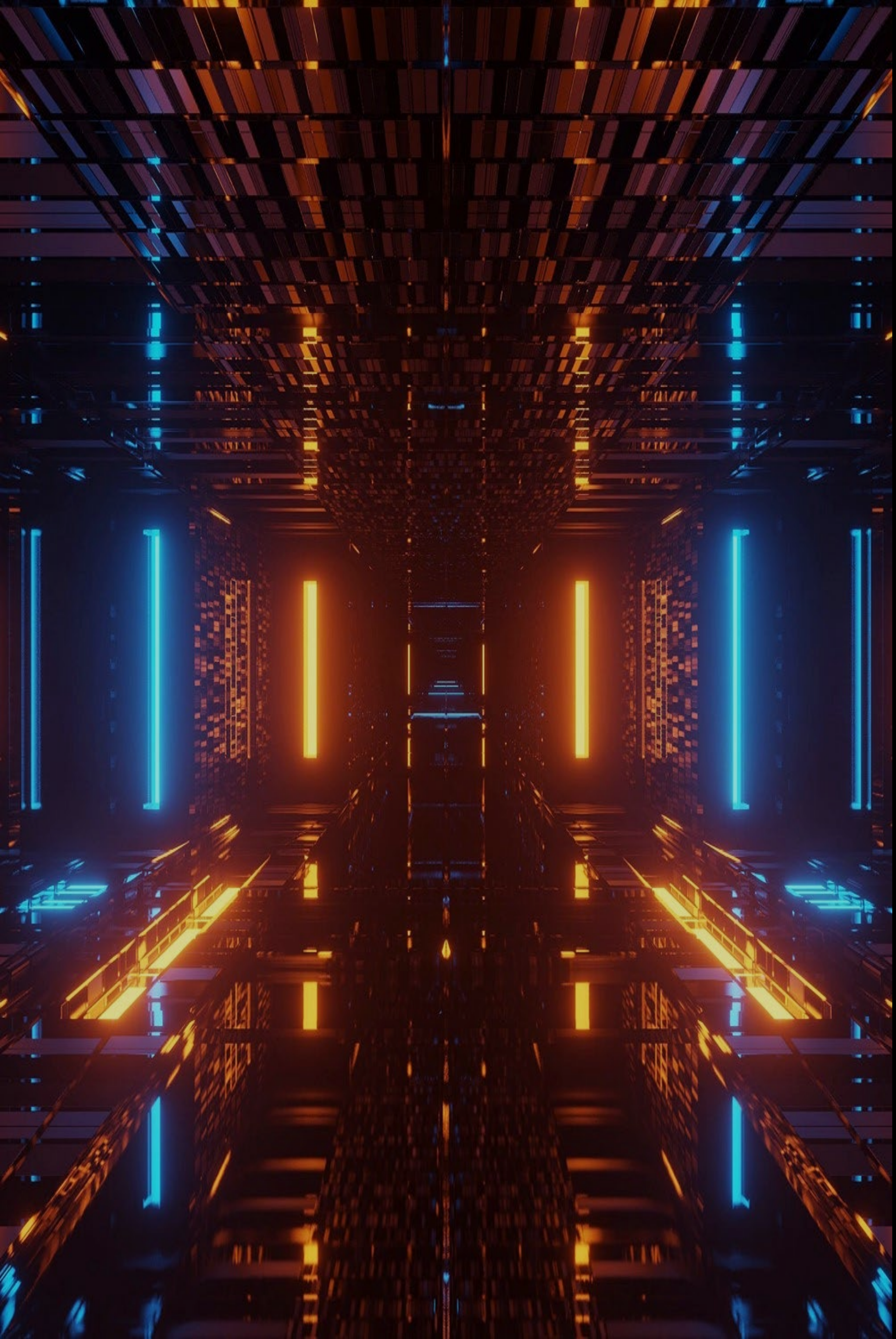
SVD can directly compute the matrix pseudoinverse for solving least-squares problems, providing stable solutions to systems of linear equations.

5.4 SVD vs PCA

	SVD	PCA
Mathematical essence	Decomposes **any matrix** as $A = U\Sigma V^T$,where U and V are orthogonal matrices and Σ is a diagonal matrix	Based on **eigen-decomposition** of the **covariance matrix**, finds the directions (principal components) of **maximum variance** in the data
Applicable objects	Can handle matrices of any shape (**non-square**, **sparse**, etc.)	Only applicable to **numerical data matrices** (not categorical matrices); requires **centering** (zero mean)
Calculation process	Decomposes the data matrix directly, **without** explicitly computing the **covariance matrix**	Requires computing the **covariance matrix** first, then performing eigen-decomposition
Dimensionality reduction method	Achieved by retaining the singular vectors corresponding to the **top k singular values**	Achieved by retaining the eigenvectors corresponding to the **top k largest eigenvalues**

5.4 SVD vs PCA: How to use SVD to implement the PCA calculation process

	SVD	PCA
Physical meaning	Reveals the intrinsic structure of the matrix; suitable for recommendation systems, image compression, etc.	Maximizes data variance; facilitates data visualization and feature extraction
Stability	More stable for high-dimensional or sparse matrices; avoids numerical instability from computing the covariance matrix	Computing the covariance matrix may introduce numerical errors, especially when the data dimensionality is high
Use cases	Recommendation systems, natural language processing (LSA), image compression, signal denoising, etc.	Exploratory data analysis, feature dimensionality reduction (e.g., image processing), data visualization, etc.
Relation ship	**PCA can be implemented via SVD: perform SVD on the centered data matrix, and its right singular vectors are the principal components of PCA**	PCA is a specific application of SVD in the field of dimensionality reduction; the two are mathematically equivalent (after data centering)



Thanks